

## Exercise\_Join\_Condition

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### Tryout > 1: Question 1

#### Problem Statement

For the invoices on which a discount of more than 12% was given, display customer name, email in lower case and invoice id.

For the given sample data, the following is the expected output.

#### Expected Output:

| CUSTNAME | EMAIL           | INVOICEID |
|----------|-----------------|-----------|
| Nancy    | nancy@gmail.com | IN101     |
| Nancy    | nancy@gmail.com | IN110     |
| Bob      | bob@gmail.com   | IN102     |
| Clara    | clara@gmail.com | IN103     |

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name | Data Type and Size | Constraint | Description |
|-------------|--------------------|------------|-------------|
|-------------|--------------------|------------|-------------|

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### Tryout > 1: Question 1

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name     | Data Type and Size | Constraint         | Description                                                       |
|-----------------|--------------------|--------------------|-------------------------------------------------------------------|
| VendorId        | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for each vendor, should start with 'V'                  |
| VendorName      | VARCHAR(30)        | NOT NULL           | name of the vendor                                                |
| VendorContactNo | NUMERIC(10)        | UNIQUE             | contact number of the vendor                                      |
| Rating          | INTEGER            | CHECK              | rating of vendor, must be in the range of 0 to 5 (both inclusive) |

Table Structure: Product (This table provides the details of all the products)

| Column Name | Data Type and Size | Constraint | Description |
|-------------|--------------------|------------|-------------|
|-------------|--------------------|------------|-------------|

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|        |         |       |                                                 |
|--------|---------|-------|-------------------------------------------------|
| Rating | INTEGER | CHECK | must be in the range of 0 to 5 (both inclusive) |
|--------|---------|-------|-------------------------------------------------|

Table Structure: Product (This table provides the details of all the products)

| Column Name | Data Type and Size | Constraint         | Description                                                     |
|-------------|--------------------|--------------------|-----------------------------------------------------------------|
| ProductId   | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for every product, should start with 'P'              |
| PName       | VARCHAR(30)        | NOT NULL           | name of the product                                             |
| Brand       | VARCHAR(20)        | NOT NULL           | brand of the product                                            |
| PCategory   | VARCHAR(20)        |                    | category of the product                                         |
| Color       | VARCHAR(10)        |                    | color of the product                                            |
| Review      | VARCHAR(10)        | CHECK              | review of the product, should be 'Bad' or 'Good' or 'Excellent' |

Table Structure: Customer (This table provides the details of all the customers)

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Table Structure: Customer (This table provides the details of all the customers)

| Column Name  | Data Type and Size | Constraint         | Description                                                      |
|--------------|--------------------|--------------------|------------------------------------------------------------------|
| CustId       | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id of the customer, must begin with character 'C'         |
| CustName     | VARCHAR(10)        | NOT NULL           | name of the customer                                             |
| Age          | NUMERIC(2)         | CHECK              | age of the customer, should be between 0 and 99 (both inclusive) |
| CustLocation | VARCHAR(20)        |                    | location, in which the customer resides                          |
| Email        | VARCHAR(30)        | UNIQUE             | email id of the customer                                         |
| Gender       | CHAR(1)            | CHECK              | gender of the customer, should be 'M' or 'F'                     |

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### Tryout > 1: Question 1

M O I F

Table Structure: VendorOffering (This table provides the details of all the products offered by different vendors)

| Column Name | Data Type and Size | Constraint            |             | Description                                                           |
|-------------|--------------------|-----------------------|-------------|-----------------------------------------------------------------------|
| VendorId    | VARCHAR(5)         | FOREIGN KEY           | PRIMARY KEY | existing VendorId present in Vendor Table                             |
| ProductId   | VARCHAR(5)         | FOREIGN KEY, NOT NULL |             | existing ProductId present in Product Table                           |
| Price       | NUMERIC(7,2)       | NOT NULL, CHECK       |             | price offered for the product by the vendor in INR, should be greater |

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|-------|--------------|-----------------|---------------------------------------------------------------------------------|
| Price | NUMERIC(7,2) | NOT NULL, CHECK | price offered for the product by the vendor in INR, should be greater than zero |
|-------|--------------|-----------------|---------------------------------------------------------------------------------|

Table Structure: Invoice (This table provides the details of all the products purchased by the customers)

| Column Name | Data Type and Size | Constraint         | Description                                                |
|-------------|--------------------|--------------------|------------------------------------------------------------|
| InvoiceId   | VARCHAR(5)         | PRIMARY KEY, CHECK | unique InvoiceId for every Invoice, should start with 'IN' |
| CustId      | VARCHAR(5)         | FOREIGN KEY        | existing CustId present in Customer Table                  |
| VendorId    | VARCHAR(5)         |                    | existing VendorId present in Vendor Table                  |

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### Tryout > 1: Question 1

|                   |              |                |                                                                  |
|-------------------|--------------|----------------|------------------------------------------------------------------|
| VendorId          | VARCHAR(5)   | FOREIGN<br>KEY | existing VendorId<br>present in Vendor<br>Table                  |
| ProductId         | VARCHAR(5)   |                | existing ProductId<br>present in Product<br>Table                |
| QuantityPurchased | NUMERIC(2)   | NOT NULL       | quantity (number<br>of products)<br>purchased by the<br>customer |
| Discount          | NUMERIC(3,1) | NOT NULL       | discount offered<br>on the purchase<br>(in percentage)           |
| PurchaseDate      | DATE         | NOT NULL       | date of purchase                                                 |

Table Data: Vendor

| VENDORID | VENDORNAME      | VENDORCONTACTNO | RATING |
|----------|-----------------|-----------------|--------|
| V101     | Wendy Creations | 8976567890      | 4      |
| V102     | CrazyNuts       | NULL            | NULL   |

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### Tryout > 1: Question 1

Table Data: Vendor

| VENDORID | VENDORNAME             | VENDORCONTACTNO | RATING |
|----------|------------------------|-----------------|--------|
| V101     | Wendy Creations        | 8976567890      | 4      |
| V102     | CrazyNuts              | NULL            | NULL   |
| V103     | FlyingHigh Retailers   | 8126567890      | 5      |
| V104     | Global Traders         | 8005432556      | 4      |
| V105     | Caretech Enterprises   | 8116567890      | 3      |
| V106     | Infinity Solutions     | 8056781234      | NULL   |
| V107     | BrightWorks Technology | 8045678902      | 3      |
| V108     | Walgreens Retailers    | 8034567891      | 4      |

Table Data: Customer

| CUSTID | CUSTNAME | AGE | CUSTLOCATION | EMAIL            | GENDER |
|--------|----------|-----|--------------|------------------|--------|
| C101   | Alfred   | 33  | Blatney      | alfred@gmail.com | F      |

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|      |                     |            |   |
|------|---------------------|------------|---|
| V108 | Walgreens Retailers | 8034567891 | 4 |
|------|---------------------|------------|---|

Table Data: Customer

| CUSTID | CUSTNAME | AGE  | CUSTLOCATION | EMAIL             | GENDER |
|--------|----------|------|--------------|-------------------|--------|
| C101   | Nancy    | 23   | Plattsburg   | nancy@gmail.com   | F      |
| C102   | Bob      | 33   | Ithaca       | bob@gmail.com     | M      |
| C103   | Clara    | NULL | Yonkers      | clara@gmail.com   | F      |
| C104   | Vincent  | 34   | Ithaca       | vincent@gmail.com | M      |
| C105   | Alex     | 27   | Cortland     | NULL              | M      |
| C106   | Peter    | 77   | Yonkers      | peter@gmail.com   | M      |
| C107   | Rose     | 54   | Ithaca       | rose@gmail.com    | F      |
| C108   | Elina    | 18   | Auburn       | elina@gmail.com   | F      |

Table Data: Product

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

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### Tryout > 1: Question 1

Table Data: Product

| PRODUCTID | PNAME             | BRAND     | PCATEGORY   | COLOR     | REVIEW    |
|-----------|-------------------|-----------|-------------|-----------|-----------|
| P101      | High Boots        | Diesel    | Shoes       | Blue      | Good      |
| P102      | Mens Formal Shirt | Terrain   | Clothing    | Turquoise | Bad       |
| P103      | Pearl Earring     | Essence   | Accessories | Yellow    | Excellent |
| P104      | Kids Crocks       | Sparx     | NULL        | Pink      | Excellent |
| P105      | Kids Shirt        | Cosmos    | Clothing    | White     | Bad       |
| P106      | Sporty Sneakers   | Reebok    | Shoes       | White     | Good      |
| P107      | Womens Jacket     | NorthFace | Clothing    | NULL      | Good      |

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### Tryout > 1: Question 1

|      |                |           |             |       |           |
|------|----------------|-----------|-------------|-------|-----------|
| P107 | Womens Jacket  | NorthFace | Clothing    | NULL  | Good      |
| P108 | Leather Wallet | Luxurio   | Accessories | Brown | Excellent |

Table Data: VendorOffering

| VENDORID | PRODUCTID | PRICE  |
|----------|-----------|--------|
| V101     | P101      | 3000.4 |
| V102     | P102      | 1200.3 |
| V102     | P104      | 1000.8 |
| V103     | P101      | 2800.1 |
| V103     | P103      | 400.2  |
| V104     | P104      | 900.1  |
| V104     | P105      | 500.8  |
| V105     | P105      | 445    |
| V107     | P107      | 90.5   |
| V108     | P106      | 699.99 |

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|      |      |        |
|------|------|--------|
| V108 | P106 | 699.99 |
|------|------|--------|

Table Data: Invoice

| INVOICEID | CUSTID | VENDORID | PRODUCTID | QUANTITYPURCHASED | DISCOUNT | PU |
|-----------|--------|----------|-----------|-------------------|----------|----|
| IN101     | C101   | V101     | P101      | 2                 | 12.5     | 20 |
| IN102     | C102   | V102     | P102      | 3                 | 13.0     | 20 |
| IN103     | C103   | V103     | P103      | 1                 | 15.2     | 20 |
| IN104     | C104   | V104     | P104      | 1                 | 10.0     | 20 |
| IN105     | C103   | V105     | P105      | 2                 | 5.0      | 20 |
| IN106     | C106   | V102     | P104      | 7                 | 10.8     | 20 |
| IN107     | C108   | V103     | P101      | 4                 | 2.0      | 20 |
| IN108     | C107   | V104     | P105      | 2                 | 3.3      | 20 |
| IN109     | C108   | V108     | P106      | 8                 | 7.0      | 20 |
| IN110     | C101   | V103     | P101      | 10                | 20.4     | 20 |

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|      |      |        |
|------|------|--------|
| V108 | P106 | 699.99 |
|------|------|--------|

Invoice

| ID | CUSTID | VENDORID | PRODUCTID | QUANTITYPURCHASED | DISCOUNT | PURCHASEDATE |
|----|--------|----------|-----------|-------------------|----------|--------------|
|    | C101   | V101     | P101      | 2                 | 12.5     | 2023-08-11   |
|    | C102   | V102     | P102      | 3                 | 13.0     | 2023-09-23   |
|    | C103   | V103     | P103      | 1                 | 15.2     | 2023-06-10   |
|    | C104   | V104     | P104      | 1                 | 10.0     | 2023-09-01   |
|    | C103   | V105     | P105      | 2                 | 5.0      | 2023-09-08   |
|    | C106   | V102     | P104      | 7                 | 10.8     | 2023-10-02   |
|    | C108   | V103     | P101      | 4                 | 2.0      | 2023-10-12   |
|    | C107   | V104     | P105      | 2                 | 3.3      | 2023-06-12   |
|    | C108   | V108     | P106      | 8                 | 7.0      | 2023-07-18   |
|    | C101   | V103     | P101      | 10                | 20.4     | 2023-08-18   |

## Exercise\_Join\_SelectiveCustomers

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### Tryout > 1: Question 1

#### Problem Statement

Display ProductId as "Product Id", PName as "Product Name" and PCategory as "Category" of all the products that are purchased by either C101 or C108.

For the given sample data, the following is the expected output.

#### Expected Output:

| Product Id | Product Name    | Category |
|------------|-----------------|----------|
| P101       | High Boots      | Shoes    |
| P101       | High Boots      | Shoes    |
| P101       | High Boots      | Shoes    |
| P106       | Sporty Sneakers | Shoes    |

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name | Data Type and Size | Constraint         | Description                                      |
|-------------|--------------------|--------------------|--------------------------------------------------|
| VendorId    | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for each vendor, should start with 'V' |
|             |                    |                    |                                                  |

## Exercise\_InnerJoin\_SelectiveAgeGroup

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### Tryout > 1: Question 1

#### Problem Statement

Display CustId as "Customer ID" and name of all customers as "Customer Name" whose age is greater than 35. Along with that display the ProductId as "Product ID" of the product(s) that the customer has purchased.

For the given sample data, the following is the expected output.

#### Sample Output:

| Customer ID | Customer Name | Product ID |
|-------------|---------------|------------|
| C106        | Peter         | P104       |
| C107        | Rose          | P105       |

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name     | Data Type and Size | Constraint         | Description                                      |
|-----------------|--------------------|--------------------|--------------------------------------------------|
| VendorId        | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for each vendor, should start with 'V' |
| VendorName      | VARCHAR(30)        | NOT NULL           | name of the vendor                               |
| VendorContactNo | NUMERIC(10)        | UNIQUE             | contact number of the                            |

## Exercise\_Join\_ConditionOnMonth

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### Tryout > 1: Question 1

#### Problem Statement

Display customer id and customer name of all the customer(s) who have purchased any product in the month of 'August'.

Hint: A customer might have purchased products more than once in the month of August. So, duplicate rows should not appear in the result.

For the given sample data, the following is the expected output.

#### Sample Output:

| CustId | CustName |
|--------|----------|
| C101   | Nancy    |

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name     | Data Type and Size | Constraint         | Description                                      |
|-----------------|--------------------|--------------------|--------------------------------------------------|
| VendorId        | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for each vendor, should start with 'V' |
| VendorName      | VARCHAR(30)        | NOT NULL           | name of the vendor                               |
| VendorContactNo | NUMERIC(10)        | UNIQUE             | contact number of the vendor                     |

## Exercise\_Join\_CategoryWiseCount

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### Tryout > 1: Question 1

#### Problem Statement

Identify the number of products purchased under each product category. Display the product category and number of products purchased under each category as "NUMBER\_OF\_PRODUCTS".

For the given sample data, the following records will feature as part of the output along with other record(s).

#### Sample Output:

| PCATEGORY | NUMBER_OF_PRODUCTS |
|-----------|--------------------|
| Shoes     | 4                  |
| NULL      | 2                  |

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name | Data Type and Size | Constraint         | Description                                      |
|-------------|--------------------|--------------------|--------------------------------------------------|
| VendorId    | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for each vendor, should start with 'V' |
| VendorName  | VARCHAR(30)        | NOT NULL           | name of the vendor                               |

## Exercise\_Join\_SelectiveColorProducts

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### Tryout > 1: Question 1

#### Problem Statement

Identify all the White or Blue color Product(s) offered by vendors. Display vendor id, product name, color and price of identified products. Round down the price values to lower integer.

For the given sample data, the following records will feature as part of the output along with other record(s).

#### Sample Output:

| VENDORID | PNAME      | COLOR | BASEPRICE |
|----------|------------|-------|-----------|
| V101     | High Boots | Blue  | 3000      |
| V103     | High Boots | Blue  | 2800      |
| V105     | Kids Shirt | White | 445       |

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name | Data Type and Size | Constraint         | Description                                      |
|-------------|--------------------|--------------------|--------------------------------------------------|
| VendorId    | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for each vendor, should start with 'V' |
| VendorName  | VARCHAR(30)        | NOT NULL           | name of the vendor                               |

## Exercise\_Join\_CategoryWiseSelectiveCount

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### Tryout > 1: Question 1

#### Problem Statement

For each product category, identify the number of products with Good review and QuantityPurchased is more than 8. Display product category as "Category" and number of such products as "Total Products".

For the given sample data, the following is the expected output.

#### Expected Output:

| Category | Total Products |
|----------|----------------|
| Shoes    | 1              |

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name     | Data Type and Size | Constraint         | Description                                      |
|-----------------|--------------------|--------------------|--------------------------------------------------|
| VendorId        | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for each vendor, should start with 'V' |
| VendorName      | VARCHAR(30)        | NOT NULL           | name of the vendor                               |
| VendorContactNo | NUMERIC(10)        | UNIQUE             | contact number of the vendor                     |
|                 |                    |                    |                                                  |

# Exercise\_Join\_SelectiveVendors

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## Tryout > 1: Question 1

### Problem Statement

For the products that are offered by vendor(s) whose name is more than 15 characters long and whose price is more than 1000, display vendor name, product id and price.

For the given sample data, the following is the expected output.

### Sample Output:

| VENDORNAME           | PRODUCTID | PRICE   |
|----------------------|-----------|---------|
| FlyingHigh Retailers | P101      | 2800.10 |

### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name     | Data Type and Size | Constraint         | Description                                            |
|-----------------|--------------------|--------------------|--------------------------------------------------------|
| VendorId        | VARCHAR(5)         | PRIMARY KEY, CHECK | unique Id for each vendor, should start with 'V'       |
| VendorName      | VARCHAR(30)        | NOT NULL           | name of the vendor                                     |
| VendorContactNo | NUMERIC(10)        | UNIQUE             | contact number of the vendor                           |
| Rating          | INTEGER            | CHECK              | rating of vendor, must be in the range of 0 to 5 (both |

Exercise\_Join\_DiscountCalculation

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Tryout > 1: Question 1

Problem Statement

For all invoices in which more than 2 products were purchased, display customer id, invoice id, actual price of product as "MRP" and price applicable after applying discount as "FINAL\_PRICE". Calculate actual price and final price, based on quantity and discount. Round off both the prices up to 0 decimal places.

For the given sample data, the following is the expected output.

**Expected Output:**

| CUSTID | INVOICEID | MRP   | FINAL_PRICE |
|--------|-----------|-------|-------------|
| C102   | IN102     | 3601  | 3133        |
| C106   | IN106     | 7006  | 6249        |
| C108   | IN107     | 11200 | 10976       |
| C101   | IN110     | 28001 | 22289       |
| C108   | IN109     | 5600  | 5208        |

Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name | Data Type and Size | Constraint | Description |
|-------------|--------------------|------------|-------------|
|             |                    |            |             |

## Exercise\_Join\_GB\_OB

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### Tryout > 1: Question 1

#### Problem Statement

Find the total number of products of each brand purchased at each location. Arrange the data in ascending order of location followed by brands in ascending order.

For the given sample data, the following rows will appear as a part of output along with other records.

#### Sample Output:

| LOCATION | BRAND   | PRODUCTSSOLD |
|----------|---------|--------------|
| Auburn   | Diesel  | 4            |
| Auburn   | Reebok  | 8            |
| Ithaca   | Cosmos  | 2            |
| Ithaca   | Sparx   | 1            |
| Ithaca   | Terrain | 3            |

#### Database Structure and Data:

Table Structure: Vendor (This table provides the details of all the vendors)

| Column Name | Data Type and Size | Constraint | Description |
|-------------|--------------------|------------|-------------|
|             |                    |            |             |

HSQL